

IMRAN AHSAN

Graph Unlearning · Explainable AI · Privacy-Preserving ML · Natural Language Processing

Postdoctoral Researcher · Chung-Ang University, South Korea

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PROFESSIONAL PROFILE

AI researcher working on efficient graph unlearning, explanation-based verification, and realistic deletion benchmarks, with a Ph.D. in Engineering from Chung-Ang University. Research connects model maintenance with privacy diagnostics and auditable evaluation. First-author contributions include Text2Forget, NodeVanisher, and Forget and Explain at top venues. Broader work spans reproducible evaluation, data quality and governance, information retrieval, and AI software engineering. Brings six years of university teaching and more than four years of software-development experience.

RESEARCH INTERESTS

- Graph Neural Networks and graph representation learning
- Explainable AI (XAI) and transparent model verification
- Graph unlearning, privacy-preserving ML, and AI regulatory compliance
- Natural Language Processing and Trustworthy AI applications

EDUCATION

PhD in Engineering | *Department of Smart City · Chung-Ang University, South Korea* *2023 – 2026*

Degree Awarded: August 2026 | **Supervisor:** Prof. Mucheol Kim | **CGPA:** 4.26 / 4.5

Thesis Title: *Toward Efficient and Effective Graph Unlearning: Challenges, Methods, and Evaluation.*

MS in Software Engineering | *National University of Sciences and Technology (NUST), Pakistan* *2014 – 2017*

Supervisor: Prof. Wasi Haider Butt | **CGPA:** 3.25 / 4.0

Thesis Title: *Automated Test Case Generation from User Specifications using NLP.*

BS in Software Engineering | *International Islamic University Islamabad (IIUI), Pakistan* | **CGPA:** 2.91 / 4.00 *2009 – 2013*

RESEARCH, ACADEMIC & INDUSTERY EXPERIENCE

Postdoctoral Researcher | *Data Intelligence Laboratory, Chung-Ang University* *Oct, 2026 - Present*

- Leads advanced research tracks in trustworthy AI, model maintenance, algorithmic governance frameworks.
- Develop methods, benchmarks, and verification protocols for graph unlearning, explainable AI, and privacy-aware model updates.
- Build reproducible Python, PyTorch, and PyTorch Geometric pipelines spanning GCN, GAT, GraphSAGE, membership-inference diagnostics, and multi-view evaluation.
- Produce peer-reviewed publications, open-source implementations, systematic reviews, and benchmark artifacts with documented provenance.

Lecturer in Computer Science | *National University of Modern Languages, Islamabad, Pakistan* *2017-2023*

- Taught Object-Oriented Programming with Java, Software Engineering, Database Management Systems and SQL, Programming Fundamentals, C#/.NET, and software-development practice.
- Designed lectures, laboratories, project-based assessments, rubrics, examinations, and formative and summative feedback.

- Supervised final-year projects and supported curriculum development and student mentoring across varied programming backgrounds.

Software Developer | *Pakistan* *4+ years*

- Developed ASP.NET-based university portals, admissions systems, student-service modules, HR systems, and business applications.
- Worked with C#, ASP.NET, JavaScript, HTML/CSS, Microsoft SQL Server, MySQL, and Oracle in database-backed application environments.

PUBLICATIONS

Published / Accepted

Journal Articles (4):

[1] **Imran Ahsan**, Brij B. Gupta, Jinsung Kim, and Mucheol Kim. "NodeVanisher: Efficient and Scalable Data Unlearning in GNNs through Targeted Subgraph Refinement." *IEEE Transactions on Consumer Electronics*, accepted, 2026.
Metrics: JCR 2025 Q1; Journal Impact Factor 9.9.

[2] **Imran Ahsan**, Muhammad Waseem Anwar, JungYoon Kim, and Mucheol Kim. "Unlocking the Power of Graph Neural Networks-A Systematic Literature Review of Application-Oriented GNN Studies." *Computer Modeling in Engineering & Sciences*, vol. 148, no. 1, Article 3, 2026. doi:10.32604/cmes.2026.080382 | artifacts
Metrics: JCR 2025 Q1 (Mathematics, Interdisciplinary Applications) and Q2 (Engineering, Multidisciplinary); Journal Impact Factor 2.9. ***Citations:*** *01*

[3] Minhyeok Choi, **Imran Ahsan**, Hyunwook Yu, Taeyoung Choe, and Mucheol Kim. "The Semantic Design Space of Retrieval-Augmented Recommender Systems: A Systematic Review of LLM-Based Approaches." *Computers, Materials & Continua*, vol. 88, no. 1, Article 3, 2026. doi:10.32604/cmc.2026.079504
Metrics: JCR 2025 Q3 (Computer Science, Information Systems; Materials Science, Multidisciplinary); Journal Impact Factor 2.4. ***Citations:*** *01*

[4] Maryum Hamdani, Wasi Haider Butt, Muhammad Waseem Anwar, **Imran Ahsan**, Farooque Azam, and Mudassar Adeel Ahmed. "A Novel Framework to Automatically Generate IFML Models From Plain Text Requirements." *IEEE Access*, vol. 7, pp. 183489-183513, 2019. doi:10.1109/ACCESS.2019.2959813
Metrics: JCR 2025 Q2 (Computer Science, Information Systems; Engineering, Electrical & Electronic; Telecommunications); Journal Impact Factor 4.2. ***Citations:*** *07*

Conferences (10):

[1] **Imran Ahsan**, Hyunwook Yu, Jinsung Kim, and Mucheol Kim. "Forget and Explain: Transparent Verification of GNN Unlearning." *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining (WSDM 2026)*, pp. 1063-1067, 2026. doi:10.1145/3773966.3779386 | code

[2] **Imran Ahsan**, Jinsung Kim, and Mucheol Kim. "From Random Forget-Sets to Realistic Natural-Language Deletion Requests." *Proceedings of The ACM Web Conference 2026 (WWW 2026)*, pp. 8741-8744, 2026.
doi:10.1145/3774904.3792954 | code

[3] **Imran Ahsan**, Hyunwook Yu, and Mucheol Kim. "GNN Unlearning Reality Checklist (GURC): A Standard for Robust, Reproducible, and Privacy-Safe Evaluation." *2025 16th International Conference on Information and Communication Technology Convergence (ICTC)*, pp. 874-876, 2025.

[4] Mudassar Adeel Ahmed, **Imran Ahsan**, Usman Qamar, and Wasi Haider Butt. "A Novel Natural Language Processing Approach to Automatically Visualize Entity-Relationship Model from Initial Software Requirements." *3rd International Conference on Communication Technologies (ComTech 2021)*, pp. 39-43, 2021. ***Citations:*** *06*
doi:10.1109/ComTech52583.2021.9616949

[5] Muhammad Waseem Anwar, **Imran Ahsan**, Farooque Azam, Wasi Haider Butt, and Muhammad Rashid. "A Natural Language Processing (NLP) Framework for Embedded Systems to Automatically Extract Verification Aspects from Textual Design Requirements." *12th International Conference on Computer and Automation Engineering (ICCAE 2020)*, pp. 7-12, 2020. ***Citations:*** *28* doi:10.1145/3384613.3384619

[6] **Imran Ahsan**, Mudassar Adeel Ahmed, Saad Rehman, Muhammad Abbas, and Muazzam Ali Khan. "A Novel NLP Application to Automatically Generate Text Extraction Concepts from Textual Descriptions." *Proceedings of the 2019*

5th International Conference on Computing and Artificial Intelligence (ICCAI 2019), pp. 55-58, 2019. ***Citations:*** *02* doi:10.1145/3330482.3330506

[7] **Imran Ahsan**, Wasi Haider Butt, Mudassar Adeel Ahmed, and Muhammad Waseem Anwar. "A Comprehensive Investigation of Natural Language Processing Techniques and Tools to Generate Automated Test Cases." *Proceedings of the Second International Conference on Internet of Things, Data and Cloud Computing (ICC 2017)*, Article 132, pp. 1-10, 2017. ***Citations:*** *48,* doi:10.1145/3018896.3036375

[8] Mudassar Adeel Ahmed, Wasi Haider Butt, **Imran Ahsan**, Muhammad Waseem Anwar, Muhammad Latif, and Farooque Azam. "A Novel Natural Language Processing (NLP) Approach to Automatically Generate Conceptual Class Model from Initial Software Requirements." *Information Science and Applications 2017*, Lecture Notes in Electrical Engineering, vol. 424, pp. 476-484, Springer, 2017. ***Citations:*** *13,* doi:10.1007/978-981-10-4154-9_55

[9] Mudassar Adeel Ahmed, **Imran Ahsan**, and Muhammad Abbas. "Systematic Literature Review: Ingenious Software Project Management while Narrowing the Impact Aspect." *Proceedings of the 2016 Conference on Research in Adaptive and Convergent Systems (RACS 2016)*, pp. 165-168, 2016. ***Citations:*** *10,* doi:10.1145/2987386.2987422

[10] Mudassar Adeel Ahmed, **Imran Ahsan**, Muazzam A. Khan Khattak, and Saad Rahman. "Management of Road Lights System Using Vehicular Ad-Hoc Networks." *2016 2nd IEEE International Conference on Computer and Communications (ICCC 2016)*, pp. 2292-2296, 2016. ***Citations:*** *01* doi:10.1109/CompComm.2016.7925108

Under Review / Revision/ In Preparation

[1] **Imran Ahsan** et al. "GDPRBench: A Controller-Side Benchmark for Accuracy-Triggered Graph Unlearning." *In preparation*, 2026.

[2] **Imran Ahsan** et al. "Retrain-Aligned Explanation Auditing for GNN Unlearning." *In preparation, 2026*.

RESEARCH CONTRIBUTIONS & OPEN SCIENCE

- Efficient graph unlearning: request-centered and subgraph-local update strategies that reduce retraining scope while preserving model utility and privacy-oriented evaluation.
- Transparent verification: explanation-shift and structural diagnostics that provide human-understandable evidence of how designated graph entities influence a model before and after unlearning.
- Realistic and auditable evaluation: natural-language erasure requests, data-quality-triggered forget sets, reproducible provenance, and multi-view assessment protocols for graph learning systems.
- Evidence synthesis and research software: systematic reviews of application-oriented GNNs and retrieval-augmented recommendation, supported by public implementations and reusable benchmark artifacts.
- **Open resources:** F-E | Text2Forget | GNN-SLR

TEACHING & SUPERVISION

- Courses: Object-Oriented Programming with Java; Software Engineering; Database Management Systems and SQL; Programming Fundamentals; C#/.NET; and software-development practice.
- Instructional design: lecture-laboratory integration, project-based assessment, rubric development, examination design, marking, and formative and summative feedback.
- Student development: final-year project supervision and mentoring in software design, database-backed applications, testing, implementation, and technical documentation.

HONORS & SCHOLARSHIPS

- Chung-Ang University Young Scientist Scholarship (CAYSS), 2023 - 2025.
- Brain Korea 21 (BK21) Scholarship, 2023 - 2026.
- Recommended by Chung-Ang University for the F-2-7S Global Talent Visa, 2026.

TECHNICAL SKILLS

Graph AI & Trustworthy ML	Graph neural networks; graph and machine unlearning; explainable AI; privacy diagnostics; responsible model updating; data-quality-aware evaluation.
ML / AI Data Science & Research	Experimental design; benchmark construction; reproducible evaluation; systematic literature reviews; natural language processing; information retrieval; retrieval-augmented systems.
Programming & ML	Python; PyTorch; PyTorch Geometric; scikit-learn; Java; C#; ASP.NET; JavaScript.
Databases & Web	SQL; Microsoft SQL Server; MySQL; Oracle; database-backed enterprise applications; HTML/CSS.

REFERENCES

1. Professor Mucheol Kim
   School of Computer Science and Engineering
   Chung-Ang University, Seoul, Republic of Korea
   Email: kimm@cau.ac.kr
   Relationship: PhD supervisor

2. Associate Professor Jinsung Kim
   School of Computer Science and Engineering
   Chung-Ang University, Seoul, Republic of Korea
   Email: kimjsung@cau.ac.kr
   Relationship: PhD Dissertation Committee Chair

Additional references are available on request.